

IFTE0007 — Decentralised Finance and Blockchain

Asset Tokenisation Design: Pokémon Card Vault Token (PCVT)

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GitHub link: <https://github.com/hoisaac484/pokemon-card-vault-token>

Smart Contract Address (Sepolia Testnet): [0xD471BF2aebF9D8c3Ff3f1Aace20fbAb95daB9E87](#)

1. Asset Description and Value Source

The asset chosen for tokenisation is a portfolio of Pokémon trading cards, graded and authenticated by recognised third-party agencies such as the Professional Sports Authenticator (PSA) or Beckett Grading Services (BGS), and stored securely in a custodial vault.

Pokémon cards derive value from three intersecting sources. Scarcity is primary: Base Set cards from 1999 were printed in limited quantities and PSA 9 or PSA 10 copies are genuinely rare (Glew, 2016). The Pokémon franchise remains one of the world's highest-grossing entertainment brands, sustaining collector demand across generations (Tobin, 2026). The grading system creates a transparent pricing hierarchy: a PSA 10 Charizard commands materially more than a PSA 7 copy, giving the asset class more valuation rigour than ungraded collectibles (PriceCharting, 2026).

Unlike equities or bonds, graded cards generate no cash flows. Returns come entirely from capital appreciation, a profile closer to fine art than conventional financial instruments. Auction data shows blue-chip graded cards appreciated substantially through the 2010s and 2020s, with select PSA 10 first-edition cards reaching six-figure valuations (Goldin, 2026).

The vault holds a diversified basket selected on grade quality (PSA 8 minimum), franchise relevance, and historical price performance. Pooling across multiple cards reduces concentration risk, while grading certification provides an externally verified valuation anchor for the token's net asset value (Theresa Sophie, 2023).

2. Rationale for Tokenisation

The graded Pokémon card market suffers from three structural inefficiencies that tokenisation directly addresses: illiquidity, access barriers, and opacity.

The primary problem is illiquidity. A single PSA 10 Charizard First Edition may fetch £50,000 or more, excluding most retail investors. Selling a physical card takes weeks. Tokenising the vault into ERC-20 shares creates a divisible, near-instantly transferable instrument that lets investors gain proportional exposure without purchasing high-value cards outright (Harris, no date).

The second problem is access barriers. The market is dominated by specialist dealers and well-connected collectors. A tokenised vault opens access to any investor who completes the platform-level verification process at a much lower minimum investment (Stohniiev, 2025).

The third problem is opacity. Physical card markets lack standardised price discovery mechanisms. Auction results are episodic and platform-specific, making it difficult to assess real-time portfolio value. By anchoring the token to an on-chain NAV variable and publishing inventory via a public metadata URI, this design introduces a layer of transparency that the physical market currently lacks (EMURGO, 2025).

Target holders include retail and semi-institutional investors seeking alternative asset exposure, collectors wanting diversified access without tying capital in a single card, and DeFi-native participants interested in real-world asset tokenisation. Primary issuance is gated by off-chain KYC verification at the platform level, consistent with UK regulatory expectations (A Data Pro, 2025). On-chain transfers are otherwise permissionless, supporting open secondary market participation.

3. Token Structure and Design

PCVT is an ERC-20 fungible token. Because tokens represent proportional participation in the entire vault rather than rights to any specific card, all tokens are economically equivalent: one PCVT carries the same pro-rata NAV claim as any other. This fungibility makes ERC-20 the correct choice over ERC-721 (NFT), which would imply unique, specific rights and misrepresent the pooled nature of the investment (Ethereum.org, 2026).

3.1 Supply Rules

The contract uses a capped adjustable supply. A `maxSupply` set as an immutable at deployment defines the absolute upper bound on tokens that can ever exist, preventing unlimited dilution. Within it, the operator mints tokens in discrete issuance rounds as new cards enter the vault. Each `mint()` call increments an `issuanceRound` counter, creating an auditable on-chain record of primary market event (Chainlink, 2025).

3.2 Rights and Transferability

Token holders hold a pro-rata economic interest in the pooled vault, benefiting from card portfolio appreciation and exiting by redeeming tokens or trading on secondary markets. Transfers are permissionless between any wallet addresses using standard ERC-20 `transfer` and `transferFrom` functions. The inherited `approve()` and `allowance()` functions enable delegated transfers, letting holders authorise DEX contracts or lending protocols to move tokens on their behalf, a key requirement for DeFi protocol integration (Buterin & Vogelsteller, 2015).

3.3 Redemption Mechanism

The `redeem()` function lets holders initiate an exit by burning their balance and emitting a `TokensRedeemed` event. Redemption is gated by a `redemptionEnabled` flag, letting the vault run in closed-end mode during card acquisition and open exits during designated windows. Settlement is off-chain: the burn signals intent, and the custodian processes the cash payout based on the current `navPerToken`.

3.4 Compliance Architecture

KYC and AML compliance is handled off-chain at the platform level rather than enforced through an on-chain whitelist. An on-chain whitelist would block tokens from reaching unapproved addresses and break DEX compatibility. Instead, the operator verifies identity before granting issuance portal access, while on-chain transfers remain permissionless. The contract cannot enforce compliance and relies on the custodian's off-chain processes, consistent with most real-world asset tokenisation platforms (A Data Pro, 2025).

3.5 Data Transparency

The `navPerToken` variable publishes the appraised value per token on-chain in GBP pence, updated via `updateNAV()` following each appraisal cycle. The `lastNAVUpdate` timestamp records when this figure was last changed, letting holders judge staleness without relying on operator communication. The `vaultMetadataURI` links to an off-chain inventory document that must include the PSA or BGS certificate number, grade, and certificate link for every card in the vault, following the standard approach used by real-world asset tokenisation platforms. The `lastMetadataUpdate` timestamp records on-chain when this document was last refreshed, providing a transparent audit trail for investors.

4. Market Access and Liquidity Design

4.1 Primary Issuance

Primary issuance occurs when the operator mints PCVT tokens against verified card deposits. The issuanceRound counter tracks each distinct minting event, marking a new primary market phase. In Round 1, an initial basket is deposited, appraised, and a corresponding quantity of tokens is distributed to early investors at a price reflecting the initial NAV. Subsequent rounds occur as the vault acquires additional cards up to the maxSupply cap (Motilal Oswal, no date).

Pricing at issuance is the aggregate appraised card value divided by tokens minted. A £100,000 vault issuing 1,000,000 PCVT tokens implies an initial NAV of £0.10 per token, recorded on-chain via navPerToken.

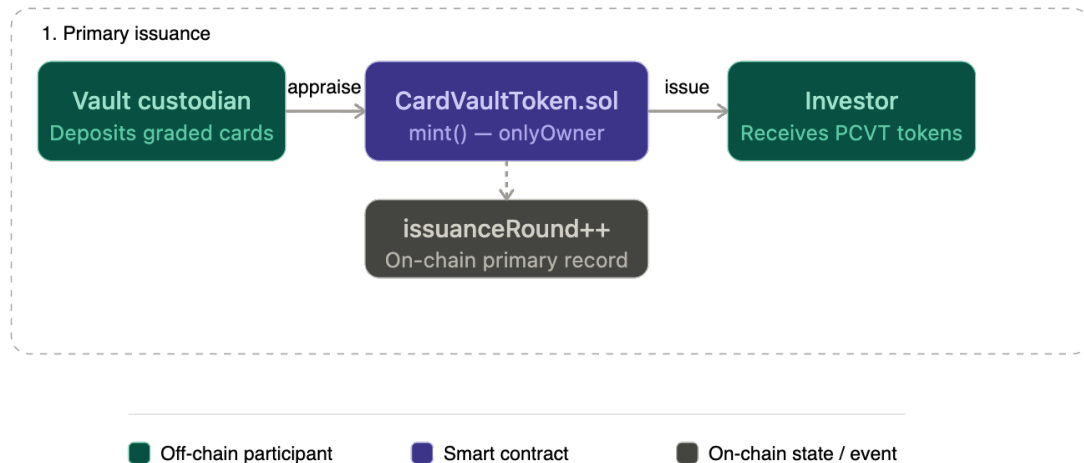
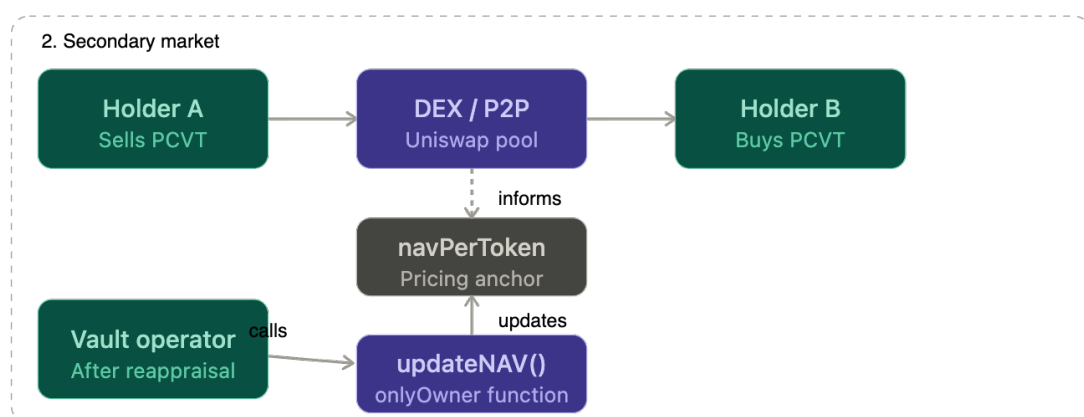


Figure 4.1 Primary Issuance Transaction Flowchart

4.2 Secondary Market

After primary issuance, PCVT tokens can be freely traded peer-to-peer or through decentralised exchanges such as Uniswap. Transfers are permissionless and fully compatible with EVM-compatible DEX infrastructure. KYC and AML obligations are met off-chain at the platform level during initial purchase.

Secondary price discovery is determined by supply and demand rather than the operator's NAV estimate. The navPerToken variable serves as a publicly verifiable pricing anchor that rational buyers and sellers can calibrate against, though it does not enforce secondary market prices. Tokens may trade at a premium to NAV if the portfolio is expected to appreciate, or at a discount if redemption demand is high or custodian confidence weakens. This premium/discount dynamic is a well-understood feature of closed-end fund structures and is a natural consequence of separating on-chain token liquidity from off-chain asset illiquidity (Ita, 2024).



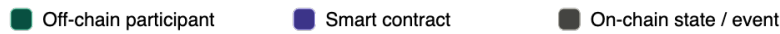


Figure 4.2 Secondary Market Transaction Flowchart

4.3 Liquidity Mechanisms

Because PCVT is a standard ERC-20, it is natively compatible with existing AMM infrastructure without requiring a bespoke exchange. The operator or early liquidity providers could seed a Uniswap v2 or v3 pool pairing PCVT with ETH or a stablecoin such as USDC (Uniswap Docs, no date). This allows continuous on-chain price discovery without any custom protocol development. The depth of the liquidity pool determines how efficiently large trades can be executed without significant price impact, which is a practical constraint given the relatively niche investor base for this asset class.

The redemptionEnabled flag provides a second liquidity channel distinct from secondary market trading. When open, holders can exit directly through the contract by burning tokens and receiving a cash payout from the custodian based on current NAV. This creates two exit routes: instant secondary market sale at the prevailing token price, or a formal redemption at NAV subject to the operator's settlement timeline. The availability of both channels improves the overall liquidity profile of the token relative to a design with only one exit mechanism (Sankey, 2024).

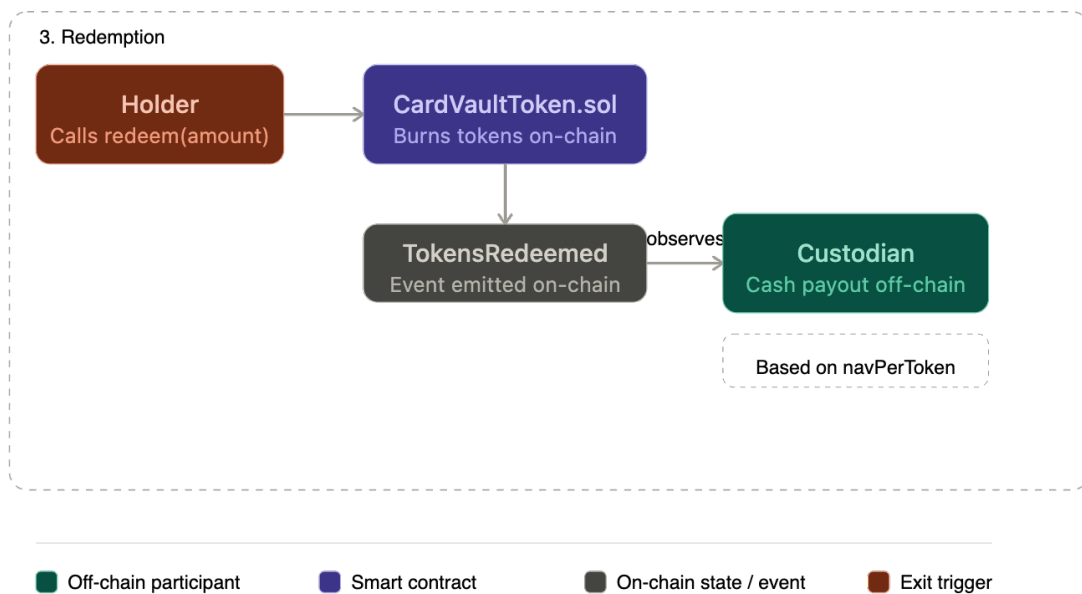


Figure 4.3 Redemption Transaction Flowchart

5. Risk Analysis and Limitations

5.1 Custody, Authenticity and Transparency Risk

The most fundamental risk in this design is that the entire value proposition depends on trusting the vault operator. There is no on-chain mechanism to verify that the cards exist, that they are stored safely, or that redemption proceeds are paid out correctly. Cards could be damaged, stolen, or mishandled without any on-chain signal to holders. This is an unavoidable limitation of tokenising physical assets, since the blockchain records claims but cannot reach into a physical vault (QuillAudits Team, 2026). In a production deployment, independent custody audits, insurance, and a multisig governance structure would partially address this, but cannot eliminate the trust dependency entirely.

Authenticity and grading risk remains material despite certificate documentation being required in the metadata URI. The contract cannot verify that the listed PSA or BGS certificates correspond to cards actually held in the vault, that grades have not been disputed or revised since certification, or that the inventory document itself is accurate. A malicious operator could list valid certificates for cards that have since been removed or substituted (Behnke, 2024). A single grade difference can materially change a card's price, and if market confidence in the vault's authentication process weakens, token value could fall sharply regardless of on-chain mechanics.

Transparency risk compounds both of the above. Holders must trust the operator's reporting on what cards are in the vault, whether they remain there, whether they are insured, and how they are valued. Two on-chain timestamps partially mitigate this: `lastNAVUpdate` records when the NAV was last set, and `lastMetadataUpdate` records when the inventory was last refreshed. Both are publicly readable, letting any holder assess reporting staleness. However, timestamps only confirm when the operator last acted, not that the underlying documentation is accurate.

5.2 Legal and Regulatory Risk

This design may meet the UK FCA's definition of a collective investment scheme, requiring authorisation that a smart contract deployment alone does not provide. Because compliance is handled off-chain, the design also depends on the operator's diligence in conducting KYC and AML checks at primary issuance. If screening fails, regulatory liability rests with the custodian rather than the protocol (CMS, 2026).

A deeper legal concern is that token holders may not hold direct legal ownership of the underlying cards. In practice, holders may only have contractual claims against the operator rather than property rights over the cards. In an insolvency scenario, contractual creditors typically rank behind secured creditors, meaning token holders could lose their economic claim even if the physical cards still exist (Ragha & Ossio, 2022). No smart contract mechanism can resolve this without legal documentation explicitly granting property rights to token holders.

5.3 Liquidity and Redemption Risk

Although PCVT tokens are freely transferable on-chain, the underlying vault assets are illiquid physical collectibles. This creates a structural liquidity mismatch: the token can be traded continuously on secondary markets, but graded Pokémon cards can only be liquidated through episodic auction or dealer sales processes that may take days or weeks. In a scenario where a significant portion of holders wish to redeem simultaneously, the vault operator may be forced to liquidate cards quickly at below-market prices, compressing the NAV per token and the cash payout available to redeeming holders (QuillAudits Team, 2026).

5.4 Valuation, Market and Concentration Risk

Card valuations are subjective and episodic. The `navPerToken` figure is set manually by the operator with no on-chain verification mechanism. Unlike liquid asset classes where oracle price feeds exist, graded collectibles have no decentralised price discovery infrastructure, making manual appraisal-based NAV updates an unavoidable design constraint rather than an oversight. If the operator overvalues the portfolio, tokens issued at that NAV will be economically diluted relative to the true asset value. The graded card market is also subject to trend-driven demand cycles. prices that appreciated sharply during 2020-2021 experienced corrections thereafter. Unlike bonds or property income, Pokémon cards generate no cash flows, so token value depends entirely on collector enthusiasm, scarcity narratives, and speculative cycles. If sentiment weakens, token value may fall sharply with no income floor to stabilise it. Holders bear this market risk in full (ChainUp, 2026).

Concentration risk is a related concern. A vault may be heavily weighted toward a few expensive cards, one generation, or one character such as Charizard. If demand shifts away from that niche, or a popular set is reprinted reducing scarcity, the portfolio could depreciate sharply. The PSA 8 minimum provides a quality floor but does not prevent narrow concentration, leaving the vault structurally fragile to sentiment shifts within the collectibles space.

5.5 Smart Contract, Centralisation and Fee Limitations

The contract builds on OpenZeppelin's audited ERC20, Ownable, and Pausable libraries, reducing implementation risk significantly. However, the custom `_update` hook and `redeem()` function have not been professionally audited; a bug in the redemption path could allow tokens to burn without triggering correct settlement. The owner's exclusive control over `mint()`, `pause()`, and `updateNAV()` means the system is not meaningfully decentralised (Ethereum.org, 2026). A compromised operator could freeze the contract, mint tokens without justification, or publish a false NAV. One practical mitigation is deploying ownership to a Gnosis Safe 2-of-3 multisig, requiring two independent signatories for any sensitive call, with no contract code changes needed.

The contract also lacks an on-chain fee mechanism. In DeFi protocols such as Uniswap and Enzyme Finance, on-chain fees are standard because value flows are fully automated. For a physical asset vault, management costs such as storage, insurance, and appraisals are more appropriately handled through the legal and custodial agreement between operator and investors. Without any explicit fee structure, there is no sustainable economic model for the operator, and informally absorbed costs could quietly erode NAV over time.

6. Role of DeFi and Blockchain Infrastructure

Blockchain infrastructure underpins every stage of this scheme's lifecycle. At deployment, the contract is permanently recorded on Ethereum as an immutable, publicly auditable record of supply rules, ownership logic, and event history, replacing the role of a central register or transfer agent.

During primary issuance, `mint()` settles token allocation instantly and transparently. The `TokensMinted` event, indexed on Etherscan, provides a public record of each issuance round. The `vaultMetadataURI` bridges on-chain token logic with off-chain asset documentation, letting any participant verify vault composition.

In secondary markets, the ERC-20 standard enables PCVT to interact natively with DeFi protocols, including Uniswap liquidity pools and Aave-style lending. The `pause()` function and `redemptionEnabled` flag let the operator manage market access across the asset lifecycle.

Redemption shows where blockchain settlement is most valuable. A holder's on-chain burn creates an immutable, timestamped record the custodian can act on without relying on intermediary systems, compressing operational friction and giving both parties a shared, unambiguous record. Collectively, the contract serves as operational infrastructure for the entire vault lifecycle: issuance, secondary trading, NAV transparency, and redemption, all governed by code on a public, permissionless blockchain.

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